PREM P S TOMAR

Senior Architect and Engineering Leader focused on High-Performing Teams, Delivery Excellence and Cross-Functional Execution

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PROFESSIONAL SUMMARY

Product engineering and platform architecture leader with 14+ years of experience building high-availability distributed systems, microservices, workflow platforms, integration-heavy products, and cloud-native backend systems across fintech, media, commerce, logistics, security, and regulated enterprise environments.

Strong fit for senior product engineering leadership requiring distributed systems design, platform consolidation, legacy modernization, product ownership, architecture governance, cloud-native engineering, REST API design, API and data-contract standardization, event-driven design, observability, reliability, and secure production operations.

Hands-on depth across Go / GoLang, Java/Spring Boot, Python/FastAPI, Rust, REST APIs, gRPC-style service contracts, Kafka/Spark-style event and data pipelines, Kubernetes, AWS, Aurora/PostgreSQL, MongoDB, Redis/Valkey, workflow orchestration, AI/ML-enabled decision support, recommendation engines, multi-modeling, probabilistic models, forensic modeling, RAG/LLM integrations, and GenAI-assisted engineering.

Known for translating ambiguous product, business, compliance, operations, and executive stakeholder expectations into scalable platform roadmaps, technical standards, release plans, architecture reviews, RCA-led operational improvements, and engineering practices that improve maintainability, throughput, resilience, and auditability while leading senior and architect-level engineers.

CORE SKILLS

Backend and Platform Stack: Go / GoLang, Java, Spring Boot, Python, FastAPI, Rust, Node.js, TypeScript, distributed systems design, microservice patterns, REST API design, API contracts, gRPC-style contracts, API gateways, service decomposition, service boundaries, event-driven architecture

Workflow and Decisioning: BPMN, Camunda, Temporal, Flowable concepts, state-machine design, workflow orchestration, intelligent routing, approval flows, SLA visibility, audit trails, case lifecycle, document-processing automation

Data and Event Platforms: Kafka-style event streaming, Spark-style data processing, PostgreSQL/Aurora, MongoDB, Elasticsearch-style search, Redis/Valkey, schema governance, data integrity, high-volume event and insight pipelines, feature-ready datasets, analytics-ready streams

Cloud, DevOps and Reliability: AWS, Kubernetes, OpenShift-aligned deployment patterns, Docker, service mesh concepts, CI/CD, Git workflows, observability, monitoring, tracing, logging, production RCA, capacity planning, 99.999%-oriented reliability practices

AI / Automation: AI/ML workflow integration, AWS Bedrock, ML-driven decisioning, automated document processing patterns, multi-LLM gateway integration, RAG services, fraud analytics, sentiment signals, recommendation engines, multi-modeling, probabilistic models, forensic modeling, GenAI-assisted development

Leadership and Governance: architecture reviews, design reviews, code reviews, microservice governance, platform standards, API governance, data schema governance, senior stakeholder management, executive communication, cross-functional governance, roadmap negotiation, decision facilitation, expectation management, engineering culture, mentoring senior and architect-level engineers, release governance, operational readiness

Security and Compliance: PCI/SOC2-aligned workflows, secure service boundaries, privacy-aware data handling, auditability, regulated enterprise delivery, risk-oriented decision support, governance-driven delivery

PROFESSIONAL EXPERIENCE

SwissClear / TAPPP | Feb 2024 - Present

Senior Architect / Principal Solutions Architect

Tech / Environment: Go / GoLang, Java, Spring Boot, Python, FastAPI, Rust, Node.js, AWS, Kubernetes, Aurora, AppSync, Valkey, Redis, Distributed Systems, Microservice Patterns, REST API Design, API Contracts, gRPC-style Contracts, Event-driven Architecture, BPMN, Camunda, Temporal, Workflow Orchestration, Decisioning Platforms, Observability, AWS Bedrock, RAG, LLMs, Fraud Detection, Decision Support APIs, PCI/SOC2

- Directed product architecture and platform engineering for distributed backend, runtime, workflow, and micro-app systems serving 7M+ daily users across high-scale client ecosystems including DTV, Globo, FIFA, Roku, BetPlay, Blackhawk Network, LiveLike, Adyen, Adidas, and Kraken.
- Drove platform consolidation by moving client-specific runtime and engagement capabilities toward reusable plug-and-play modules, shared service contracts, REST API standards, common onboarding flows, and configurable decision/workflow patterns.
- Designed workflow and case-management style orchestration patterns using BPMN, Camunda, and Temporal for intelligent routing, state transitions, approvals, event-driven execution, SLA visibility, and audit-friendly operational trails.
- Led Java-to-Go modernization for performance-sensitive services after profiling CPU, memory, latency, throughput, and data-access bottlenecks; improved API response times through asynchronous processing, network redesign, private AWS networking, Aurora, AppSync, and Valkey.
- Established architecture governance across microservice patterns, REST APIs, gRPC-style service contracts, API versioning, data schemas, event interfaces, backward compatibility, observability, deployment readiness, and PCI/SOC2-aligned secure workflows.

- Integrated AI/ML and automation capabilities using AWS Bedrock, RAG/LLM patterns, fraud-detection signals, live-feed sentiment analysis, recommendation signals, multi-model decision support, probabilistic and forensic modeling patterns, and diagnostics APIs for operational intelligence.
- Built Rust workers for high-volume event and insight payload processing, transforming production telemetry and engagement data into analytics-ready streams for big-data pipelines and downstream insight platforms.
- Led and mentored senior and architect-level engineers through architecture reviews, code reviews, technical decisioning, release planning, production RCA, and platform governance while managing product, leadership, QA, infrastructure, DevOps/SRE, compliance, operations, and client-facing stakeholder expectations.

Link to Any | May 2023 - Feb 2024

Engineering Mentor / Senior Software Engineer

Tech / Environment: Go / GoLang, Java, Python, FastAPI, Node.js, AWS, REST API Design, API Contracts, Microservice Patterns, Integration Platforms, Data Migrations, CI/CD, Observability, Partner APIs, Operational Dashboards

- Built end-to-end backend systems and internal engineering tools using distributed service patterns for data migrations, partner integrations, workflow automation, observability, QA traceability, and operational dashboards.
- Designed REST API integration services for GoDaddy, Shopify, Clover, Square, Promenade, and LightSpeed with emphasis on API contracts, data integrity, backward compatibility, partner contracts, and reliable production operations.
- Modernized selected Java/Node.js components to Go / GoLang for lower runtime overhead, stronger concurrency control, and improved CPU/memory behavior in high-volume integration jobs.
- Improved operational quality through structured logging, dashboarding, QA automation, CI/CD governance, Git workflows, review discipline, RCA-driven long-term fixes, and stakeholder-facing operational communication.
- Mentored engineers through design reviews, code reviews, debugging sessions, implementation reviews, release planning, and production-readiness practices.

MavQ | Aug 2021 - May 2023

Software Development Lead

Tech / Environment: Java, Python, FastAPI, Node.js, AWS, AWS Connect, API-first Architecture, REST API Design, Microservice Patterns, Workflow Automation, No-code Platforms, Intelligent Document Processing, EDA, Recommendation Engines, Multi-modeling, CI/CD, Product Delivery

- Led product delivery for no-code automation and workflow-oriented platform capabilities using API-first architecture, reusable abstractions, backend services, REST API patterns, recommendation-oriented data flows, and integration patterns designed for scalable execution.
- Owned complex modules from design through production support, including service design, data modeling, testing strategy, CI/CD, release planning, and operational handover.
- Created reusable database, microservice, and platform abstractions that reduced service rollout time and improved engineering consistency across product teams.
- Delivered Customer Experience Center capabilities using AWS Connect and contributed to Intelligent Document Processing automation, aligning backend systems with product, operations, support, and leadership stakeholders.
- Mentored engineers on API design, testing, automation, code quality, delivery discipline, and production readiness while driving product, QA, DevOps, operations, and cross-functional stakeholder alignment.

Swift | Nov 2019 - Aug 2021

Platform Lead / Senior Software Engineer

Tech / Environment: Go / GoLang, Java, Python, Rust, Node.js, Angular, AWS, MySQL, MongoDB, Distributed Systems, Microservice Patterns, REST APIs, Checkout, Payments, Courier Orchestration, Payment Orchestration, SenseEngine, Recommendation Engines, ML Data Processing

- Led the e-commerce platform stack covering checkout, payments, payment orchestration, courier orchestration, merchant workflows, partner integrations, SenseEngine, recommendation-oriented decisioning, and shipment-intelligence capabilities.
- Designed Java and Go-backed microservices across checkout, logistics, payment, courier, and partner systems with emphasis on microservice patterns, resiliency, data integrity, backward compatibility, and operational support.
- Built BYOC courier orchestration and BYOP payment orchestration platforms, creating workflow-style routing and decisioning patterns across merchant, logistics, and partner ecosystems.
- Implemented Rust-backed high-volume shipment data processing paths feeding ML models, improving throughput, reliability, and control over large-scale logistics datasets.
- Created SenseEngine, supporting shipment intelligence, recommendation-style decisioning, and ML-driven optimization across 300K+ monthly orders while reducing cost per shipment through optimization logic.
- Partnered with product, operations, support, engineering, partner, and leadership stakeholders to improve release quality, production reliability, incident response, and platform scalability.

D-SYS Data Solutions | Gwalior, MP | Dec 2014 - Nov 2019

Senior Full-Stack Developer | Mar 2018 - Nov 2019

Tech / Environment: Python, Data Analysis, SQL, .NET MVC, C#, Entity Framework, Xamarin, Web APIs, Transportation / E-Way Billing Systems, Compliance Workflows

- Delivered enterprise full-stack systems with .NET MVC, C#, SQL, and Xamarin, including transportation and e-way billing workflows for national road-traffic compliance use cases.
- Expanded Python use into data analysis, structured data preparation, reporting, and model experimentation for analytics-heavy product work.

- Built backend foundations for safety/security product workflows using Node.js and Express.js, guiding Angular, Node.js, Android, and backend engineers.
- Extended Python, Go, and Rust-backed engineering into ML data collection, exploratory analysis, semi-supervised learning concepts, sentiment signals, recommendation logic, and multi-model experiments.
- Contributed to smart security systems with ML data collection and security-domain intelligence to improve operational performance.
- Owned backend and application features from design through implementation, debugging, deployment support, and client-facing resolution.

Full-Stack / Mobile Developer | **Dec 2014 - Feb 2018**

Tech / Environment: Java, Java OOP, Python/Django, .NET MVC, Entity Framework, Android, iOS, MySQL, MongoDB, AWS, Linux, P2P Server Management

- Delivered mobile, backend, marketplace, database, and production-server capabilities for client-facing product and enterprise engagements.
- Engineered Java-led client systems with Python/Django support for admin workflows, APIs, client portals, and database-backed applications.
- Implemented Android/iOS applications, server-side components, P2P server support, .NET MVC marketplace features, database designs, and operational deployment workflows.
- Supported D-SYS Data Solutions product and client-delivery initiatives across mobile, marketplace, database-backed workflow, and server operations work.
- Coordinated with distributed designers, developers, marketing teams, and business stakeholders to move requirements from concept to working production applications.

Aadi Infotech Pvt. Ltd. | Gwalior, MP | Feb 2010 - Nov 2014

Software Developer

Tech / Environment: Java, Java OOP, MySQL, Apache, Linux, Python/Django, C#/.NET, Android, iOS, PHP, Web Applications, Enterprise Applications

- Started as an intern/associate during academic years and grew into a software developer role, contributing to transportation, education, social/image-editing, mobile, and enterprise applications.
- Worked with small engineering teams to design and implement MTI Transport, JOS, Academia, Party Smart, image capture/editing workflows, and customer-facing web enhancements.
- Built primarily Java-backed web and enterprise systems, with Python/Django supporting admin workflows, APIs, internal tools, and database-backed modules.
- Participated in redesigning standalone applications into more maintainable object-oriented and thin-client architectures using Java and relational databases.
- Collaborated directly with clients and business stakeholders to analyze issues, support delivery, improve usability, and contribute to product growth.

ADDITIONAL CONSULTING EXPOSURE

Freelance / Consulting Engagements | Dec 2014 - Nov 2019

- Brief concurrent consulting exposure across Millennial Group Inc., Cloud Digital Arts, Cargo 3 IVS / FindACargo, and independent client delivery, covering mobile apps, marketplace features, database-backed workflows, server operations, and release coordination.

EDUCATION

PG Program in Data Science and AI/ML - Great Learning | B.E. Information Technology - RGPV